第二章　行列式

§2.2　行列式的性质

记：

$$D=\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}.$$

将该行列式的行与列互换，不改变它们的先后顺序得到的新行列式称为 D 的**转置行列式**，记为 D^T，即：

$$D^T=\begin{vmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ \vdots & \vdots & & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{vmatrix}.$$

性质1 行列式与它的转置行列式相等，即 $\boldsymbol{D}=\boldsymbol{D}^{\mathrm{T}}$.

例
$$\begin{vmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a_{22} & a_{23} & \cdots & a_{2n} \\ 0 & 0 & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{vmatrix} = \begin{vmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ a_{12} & a_{22} & 0 & \cdots & 0 \\ a_{13} & a_{23} & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & a_{3n} & \cdots & a_{nn} \end{vmatrix}$$

$$= a_{11}a_{22}\cdots a_{nn}$$

性质1说明行列式中行与列的地位是对称的, 凡是对行成立的性质对列也同样成立.

性质2 行列式的某一行（列）元素的公因子可以提到行列式符号的外面，即

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ ka_{i1} & ka_{i2} & \cdots & ka_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = k \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

推论 若行列式中某一行（列）的元素全为零，则行列式等于零.

性质3 互换行列式的两行（列），行列式变号，即

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{kn} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = - \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{kn} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

推论 若行列式中有两行（列）的对应元素相同或成比例，则行列式等于零。

性质4 若行列式的某一行(列)的各元素都写成两项之和，则此行列式可写成两个行列式之和，即

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1}+a'_{i1} & a_{i2}+a'_{i2} & \cdots & a_{in}+a'_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

$$= \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a'_{i1} & a'_{i2} & \cdots & a'_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

性质5 将行列式的某一行(列)的各元素的k倍加到另一行(列)对应的元素上去，行列式不变.

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{j1} & a_{j2} & \cdots & a_{jn} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} \xlongequal{r_i + kr_j} \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1}+ka_{j1} & a_{i2}+ka_{j2} & \cdots & a_{in}+ka_{jn} \\ \vdots & \vdots & & \vdots \\ a_{j1} & a_{j2} & \cdots & a_{jn} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

小结 行列式的基本运算性质

- 转置性质；
- 拆项性质；
- 三个初等变换性质；

注：(1) 如果方阵A经过有限次初等变换得到矩阵B,则

$$|B| = l|A|,$$

其中l是数域$\mathbf{P}$中非零的数。

(2) 即A和B的行列式是否为零的性质不变.

例1 计算4阶行列式

$$D=\begin{vmatrix} a^2+\dfrac{1}{a^2} & a & \dfrac{1}{a} & 1 \\ b^2+\dfrac{1}{b^2} & b & \dfrac{1}{b} & 1 \\ c^2+\dfrac{1}{c^2} & c & \dfrac{1}{c} & 1 \\ d^2+\dfrac{1}{d^2} & d & \dfrac{1}{d} & 1 \end{vmatrix} \qquad (\text{已知 } abcd=1)$$

解

$$D=\begin{vmatrix} a^2 & a & \frac{1}{a} & 1 \\ b^2 & b & \frac{1}{b} & 1 \\ c^2 & c & \frac{1}{c} & 1 \\ d^2 & d & \frac{1}{d} & 1 \end{vmatrix} + \begin{vmatrix} \frac{1}{a^2} & a & \frac{1}{a} & 1 \\ \frac{1}{b^2} & b & \frac{1}{b} & 1 \\ \frac{1}{c^2} & c & \frac{1}{c} & 1 \\ \frac{1}{d^2} & d & \frac{1}{d} & 1 \end{vmatrix}$$

$$= abcd \begin{vmatrix} a & 1 & \frac{1}{a^2} & \frac{1}{a} \\ b & 1 & \frac{1}{b^2} & \frac{1}{b} \\ c & 1 & \frac{1}{c^2} & \frac{1}{c} \\ d & 1 & \frac{1}{d^2} & \frac{1}{d} \end{vmatrix} + \begin{vmatrix} \frac{1}{a^2} & a & \frac{1}{a} & 1 \\ \frac{1}{b^2} & b & \frac{1}{b} & 1 \\ \frac{1}{c^2} & c & \frac{1}{c} & 1 \\ \frac{1}{d^2} & d & \frac{1}{d} & 1 \end{vmatrix}$$

$$= abcd \begin{vmatrix} a & 1 & \frac{1}{a^2} & \frac{1}{a} \\ b & 1 & \frac{1}{b^2} & \frac{1}{b} \\ c & 1 & \frac{1}{c^2} & \frac{1}{c} \\ d & 1 & \frac{1}{d^2} & \frac{1}{d} \end{vmatrix} + (-1)^1 \begin{vmatrix} \frac{1}{a^2} & a & \frac{1}{a} & 1 \\ \frac{1}{b^2} & b & \frac{1}{b} & 1 \\ \frac{1}{c^2} & c & \frac{1}{c} & 1 \\ \frac{1}{d^2} & d & \frac{1}{d} & 1 \end{vmatrix}$$

$$= abcd \begin{vmatrix} a & 1 & \frac{1}{a^2} & \frac{1}{a} \\ b & 1 & \frac{1}{b^2} & \frac{1}{b} \\ c & 1 & \frac{1}{c^2} & \frac{1}{c} \\ d & 1 & \frac{1}{d^2} & \frac{1}{d} \end{vmatrix} + (-1)^2 \begin{vmatrix} a & \frac{1}{a^2} & \frac{1}{a} & 1 \\ b & \frac{1}{b^2} & \frac{1}{b} & 1 \\ c & \frac{1}{c^2} & \frac{1}{c} & 1 \\ d & \frac{1}{d^2} & \frac{1}{d} & 1 \end{vmatrix}$$

$$= abcd \begin{vmatrix} a & 1 & \frac{1}{a^2} & \frac{1}{a} \\ b & 1 & \frac{1}{b^2} & \frac{1}{b} \\ c & 1 & \frac{1}{c^2} & \frac{1}{c} \\ d & 1 & \frac{1}{d^2} & \frac{1}{d} \end{vmatrix} + (-1)^3 \begin{vmatrix} a & \frac{1}{a} & \frac{1}{a^2} & 1 \\ b & \frac{1}{b} & \frac{1}{b^2} & 1 \\ c & \frac{1}{c} & \frac{1}{c^2} & 1 \\ d & \frac{1}{d} & \frac{1}{d^2} & 1 \end{vmatrix}$$

$$= abcd \begin{vmatrix} a & 1 & \frac{1}{a^2} & \frac{1}{a} \\ b & 1 & \frac{1}{b^2} & \frac{1}{b} \\ c & 1 & \frac{1}{c^2} & \frac{1}{c} \\ d & 1 & \frac{1}{d^2} & \frac{1}{d} \end{vmatrix} + (-1)^3 \begin{vmatrix} a & 1 & \frac{1}{a^2} & \frac{1}{a} \\ b & 1 & \frac{1}{b^2} & \frac{1}{b} \\ c & 1 & \frac{1}{c^2} & \frac{1}{c} \\ d & 1 & \frac{1}{d^2} & \frac{1}{d} \end{vmatrix}$$

$$= 0.$$

(由 $abcd = 1$)

例2

$$\begin{vmatrix} a^2+b^2 & ac & bc \\ ac & b^2+c^2 & ab \\ bc & ab & c^2+a^2 \end{vmatrix}$$

$$= \begin{vmatrix} a^2 & ac & bc \\ ac & b^2+c^2 & ab \\ 0 & ab & c^2+a^2 \end{vmatrix} + \begin{vmatrix} b^2 & ac & bc \\ 0 & b^2+c^2 & ab \\ bc & ab & c^2+a^2 \end{vmatrix}$$

$$
\begin{aligned}
&= a\begin{vmatrix} a & ac & bc \\ c & b^2+c^2 & ab \\ 0 & ab & c^2+a^2 \end{vmatrix} + b\begin{vmatrix} b & ac & bc \\ 0 & b^2+c^2 & ab \\ c & ab & c^2+a^2 \end{vmatrix} \\
&= a\begin{vmatrix} a & 0 & bc \\ c & b^2 & ab \\ 0 & ab & c^2+a^2 \end{vmatrix} + b\begin{vmatrix} b & ac & 0 \\ 0 & b^2+c^2 & ab \\ c & ab & a^2 \end{vmatrix} \\
&= ab\begin{vmatrix} a & 0 & bc \\ c & b & ab \\ 0 & a & c^2+a^2 \end{vmatrix} + ab\begin{vmatrix} b & ac & 0 \\ 0 & b^2+c^2 & b \\ c & ab & a \end{vmatrix}
\end{aligned}
$$

$$= ab\begin{vmatrix} a & 0 & bc \\ c & b & 0 \\ 0 & a & c^2 \end{vmatrix} + ab\begin{vmatrix} b & ac & 0 \\ 0 & c^2 & b \\ c & 0 & a \end{vmatrix}$$

$$= abc\begin{vmatrix} a & 0 & b \\ c & b & 0 \\ 0 & a & c \end{vmatrix} + abc\begin{vmatrix} b & a & 0 \\ 0 & c & b \\ c & 0 & a \end{vmatrix}$$

$$= 2abc(abc + abc) = 4a^2b^2c^2$$

三角化方法

1、 低阶数字行列式

$$\begin{vmatrix} -2 & 5 & -1 & 3 \\ 1 & -9 & 13 & 7 \\ 3 & -1 & 5 & -5 \\ 2 & 8 & -7 & -10 \end{vmatrix} \overset{r_2 \leftrightarrow r_1}{=\!=} - \begin{vmatrix} 1 & -9 & 13 & 7 \\ -2 & 5 & -1 & 3 \\ 3 & -1 & 5 & -5 \\ 2 & 8 & -7 & -10 \end{vmatrix}$$

$$
-\begin{vmatrix} 1 & -9 & 13 & 7 \\ -2 & 5 & -1 & 3 \\ 3 & -1 & 5 & -5 \\ 2 & 8 & -7 & -10 \end{vmatrix}
\begin{matrix} r_2 + 2r_1 \\ r_3 - 3r_1 \\ \underline{r_4 - 2r_1} \end{matrix}
-\begin{vmatrix} 1 & -9 & 13 & 7 \\ 0 & -13 & 25 & 17 \\ 0 & 26 & -34 & -26 \\ 0 & 26 & -33 & -24 \end{vmatrix}
$$

$$
\begin{matrix} r_3 - r_4 \\ \underline{r_4 + 2r_2} \end{matrix}
-\begin{vmatrix} 1 & -9 & 13 & 7 \\ 0 & -13 & 25 & 17 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 17 & 10 \end{vmatrix}
\underline{r_4 + 17r_3}
-\begin{vmatrix} 1 & -9 & 13 & 7 \\ 0 & -13 & 25 & 17 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & -24 \end{vmatrix}
$$

$$
= -1 \times (-13) \times (-1) \times (-24) = 312
$$

2、箭形行列式（三角化）

例2.2.10　计算 $n+1$ 阶箭形行列式

$$D=\begin{vmatrix} a_0 & 1 & 1 & \cdots & 1 \\ 1 & a_1 & & & \\ 1 & & a_2 & & \\ \vdots & & & \ddots & \\ 1 & & & & a_n \end{vmatrix},$$

$$a_i \neq 0, i=0,1,2,\cdots,n.$$

$$\begin{vmatrix} a_0 & 1 & 1 & \cdots & 1 \\ 1 & a_1 & & & \\ 1 & & a_2 & & \\ \vdots & & & \ddots & \\ 1 & & & & a_n \end{vmatrix} \xrightarrow{c_1 - \sum_{j=1}^{n} \frac{1}{a_j} c_{j+1}} \begin{vmatrix} a_0 - \sum_{j=1}^{n} \frac{1}{a_j} & 1 & 1 & \cdots & 1 \\ 0 & a_1 & & & \\ 0 & & a_2 & & \\ \vdots & & & \ddots & \\ 0 & & & & a_n \end{vmatrix}$$

$$= (a_0 - \sum_{j=1}^{n} \frac{1}{a_j}) \prod_{i=1}^{n} a_i$$

同样，形如$|\searrow|$,$|\swarrow|$,$|\nearrow|$的行列式也称为箭形行列式，均可用类似的方法化为三角行列式进行计算.

3、可化为箭形的行列式

例2.2.11计算n 阶行列式

$$D=\begin{vmatrix} a_1+b_1 & b_2 & \cdots & b_n \\ b_1 & a_2+b_2 & \cdots & b_n \\ \vdots & \vdots & & \vdots \\ b_1 & b_2 & \cdots & a_n+b_n \end{vmatrix},$$

$$a_i \neq 0, \quad i=1,2,\cdots,n$$

特点 各行(列)相近

方法 各行(列)都减某一行(列),化为箭形.

$$\begin{vmatrix} a_1+b_1 & b_2 & \cdots & b_n \\ b_1 & a_2+b_2 & \cdots & b_n \\ \vdots & \vdots & & \vdots \\ b_1 & b_2 & \cdots & a_n+b_n \end{vmatrix} \xrightarrow[i\geq 2]{\boldsymbol{r}_i-\boldsymbol{r}_1} \begin{vmatrix} a_1+b_1 & b_2 & \cdots & b_n \\ -a_1 & a_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ -a_1 & 0 & \cdots & a_n \end{vmatrix}$$

$$\begin{vmatrix} a_1+b_1 & b_2 & \cdots & b_n \\ -a_1 & a_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ -a_1 & 0 & \cdots & a_n \end{vmatrix} \overset{c_1+\frac{c_i}{a_i}\boldsymbol{a}_1}{\underset{i\geq 2}{=\!=\!=}} \begin{vmatrix} a_1+b_1+a_1\sum_{i=2}^{n}\frac{b_i}{a_i} & b_2 & \cdots & b_n \\ 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & a_n \end{vmatrix}$$

$$= (a_1+b_1+a_1\sum_{i=2}^{n}\frac{b_i}{a_i})a_2a_3\cdots a_n$$

$$= (1+\sum_{i=1}^{n}\frac{b_i}{a_i})\prod_{i=1}^{n}a_i$$

4、 行（列）和相等的行列式

例 计算 n 阶行列式

$$D_n=\begin{vmatrix} 1+a_1 & a_2 & \cdots & a_n \\ a_1 & 1+a_2 & \cdots & a_n \\ \vdots & \vdots & & \vdots \\ a_1 & a_2 & \cdots & 1+a_n \end{vmatrix}$$

特点　行列式的每一行元素之和相等

方法　各列都加到同一列(例如第1列)上，再三角化.

$$D_n \overset{\substack{c_1+c_j \\ =}}{\underset{j=2,\cdots,n}{}} \begin{vmatrix} 1+\sum_{i=1}^{n} a_i & a_2 & \cdots & a_n \\ 1+\sum_{i=1}^{n} a_i & 1+a_2 & \cdots & a_n \\ \vdots & \vdots & & \vdots \\ 1+\sum_{i=1}^{n} a_i & a_2 & \cdots & 1+a_n \end{vmatrix}$$

$$
\begin{vmatrix}
1+\sum_{i=1}^{n} a_i & a_2 & \cdots & a_n \\
0 & 1 & \cdots & 0 \\
\vdots & \vdots & & \vdots \\
0 & 0 & \cdots & 1
\end{vmatrix}
\overset{\substack{r_i - r_1 \\ i=2,\cdots,n}}{=}
$$

$$
= 1+\sum_{i=1}^{n} a_i
$$

方法2 此行列式的各行相近，因此可化为箭形.

$$D_n=\begin{vmatrix} 1+a_1 & a_2 & \cdots & a_n \\ a_1 & 1+a_2 & \cdots & a_n \\ \vdots & \vdots & & \vdots \\ a_1 & a_2 & \cdots & 1+a_n \end{vmatrix}$$

$$\underset{i=2,\cdots,n}{\overset{r_i-r_1}{=}}\begin{vmatrix} 1+a_1 & a_2 & \cdots & a_n \\ -1 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ -1 & 0 & \cdots & 1 \end{vmatrix}$$

$$\overset{c_1+c_j}{\underset{j=2,\cdots,n}{=}}\begin{vmatrix} 1+\sum_{i=1}^{n} a_i & a_2 & \cdots & a_n \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{vmatrix} = 1+\sum_{i=1}^{n} a_i$$